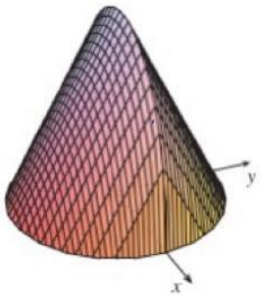
Lecture 13, July 13th

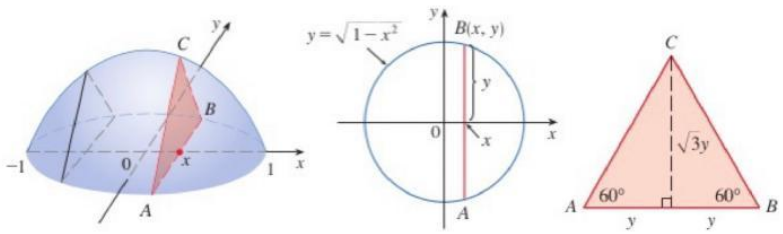
Math 225- Summer

Finding Volume using Cross-Sectional Areas:

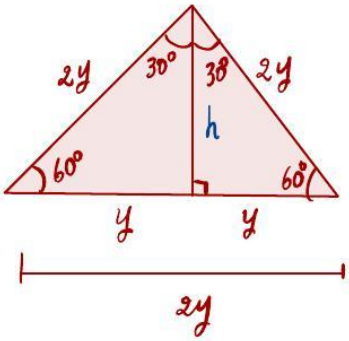
Example: this picture shows a solid with a circular base of radius 1 parallel cross-sections perpendicular to the base are equilateral triangles. Find the volume of the solid.



Sln: Let's take the circle $x^2+y^2=1$.



(a) The solid (b) Its base (c) A cross-section



$(2y)^2 = y^2 + h^2$

$4y^2 - y^2 = h^2$

$3y^2 = h^2$

$h = \sqrt{3}\,y$

So,

$A(x) = \frac{1}{2}(2y)(\sqrt{3}\,y)$

$= \sqrt{3}\,y^2$.

Since $y^2 = 1-x^2$ then

$$A(x) = \sqrt{3}(1-x^2)$$

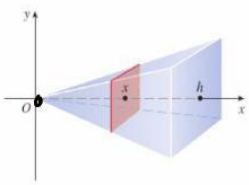
So,

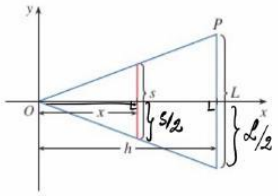
$$V = \int_{-1}^{1} A(x)\,dx = \int_{-1}^{1} \sqrt{3}(1-x^2)\,dx = 2\sqrt{3}\int_{0}^{1}(1-x^2)\,dx$$

$$= 2\sqrt{3}\left(x - \frac{x^3}{3}\,\Big|_0^1\right) = 2\sqrt{3}\left(1-\frac{1}{3}\right) = 4\frac{\sqrt{3}}{3}.$$

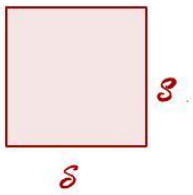
Example. Find the volume of a pyramid whose base is a square with side L and whose hight is h.

Sln.





$$\frac{x}{h} = \frac{s/2}{\ell/2} = \frac{s}{\ell} \quad \text{then}$$



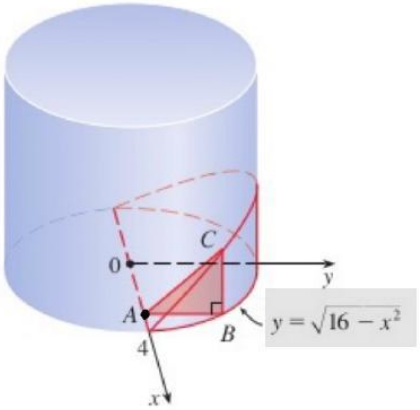
$$s = \frac{\ell \cdot x}{h}.$$

$$A(x) = s^2 = \left(\frac{\ell x}{h}\right)^2 = \frac{\ell^2}{h^2} \cdot x^2. \quad \text{then}$$

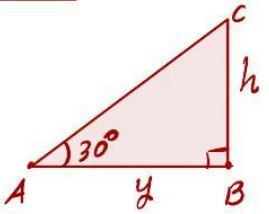
$$V = \int_0^h A(x)\,dx = \int_0^h \frac{\ell^2}{h^2} x^2\,dx = \frac{\ell^2}{h^2}\int_0^h x^2\,dx$$

$$= \frac{\ell^2}{h^2}\left\{\frac{x^3}{3}\Big|_0^h\right\} = \frac{\ell^2 h^3}{3h^2} = \frac{1}{3}\ell^2 h.$$

Example. A wedge is cut out of a circular cylinder of radius 4 by two planes. One plane is perpendicular to the axis of the cylinder the other plane intersects the first at an angle of 30° along the diameter of cylinder. Find the volume of the wedge.



Sln. Notice that this triangle:



$$A(x) = \frac{1}{2} y \cdot h.$$

$$A(x) = \frac{1}{2}\sqrt{16 - x^2} \cdot h.$$

$$\tan 30° = \frac{h}{y} \Rightarrow h = y \cdot \tan 30° = \frac{y}{\sqrt{3}} = \frac{\sqrt{16 - x^2}}{\sqrt{3}}.$$

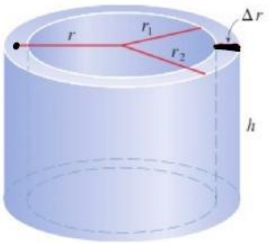
$$A(x) = \frac{1}{2}\sqrt{16 - x^2} \cdot \frac{\sqrt{16 - x^2}}{\sqrt{3}}$$

$$A(x) = \frac{16 - x^2}{2\sqrt{3}}.$$

$$V = \int_{-4}^{4} \frac{16 - x^2}{2\sqrt{3}}\,dx = \frac{2}{2\sqrt{3}}\int_0^4 (16 - x^2)\,dx = \frac{1}{\sqrt{3}}\left\{16x - \frac{x^3}{3}\Big|_0^4\right\} = \frac{128}{3\sqrt{3}}.$$

5.3 Volumes by cylindrical shells.

* The method of cylindrical shells.



Notice that the volume of this solid.

$$V = V_2 - V_1$$

$$V = \pi r_2^2 \cdot h - \pi r_1^2 \cdot h$$

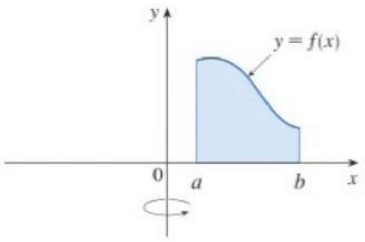
$$V = \pi (r_2^2 - r_1^2) \cdot h$$

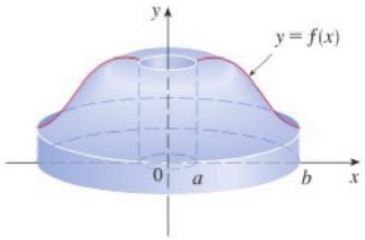
$$V = 2\pi \left(\frac{r_2 + r_1}{2}\right)(r_2 - r_1) \cdot h$$

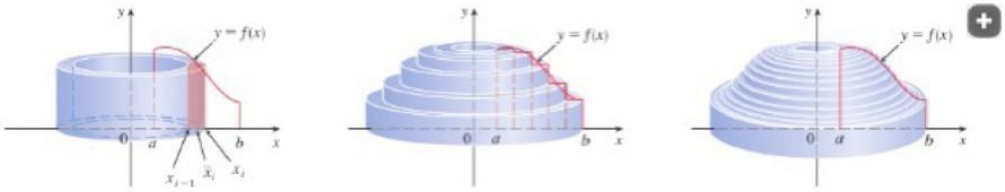
Notice that $\Delta r = r_2 - r_1$ and $r = \frac{r_1 + r_2}{2}$ so.

$$V = 2\pi r h \Delta r.$$

Now, let S be the solid obtained by rotating about y-axis the region bounded by $y = f(x)$ $\{f(x) \geq 0\}$ $y = 0$, $x = a$, $x = b$.



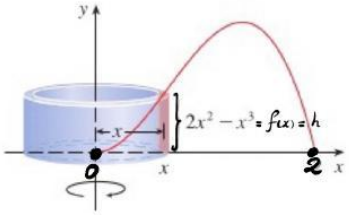




Def. The volume of the solid above, obtained by rotating about the y-axis the region under the curve $y = f(x)$ from a to b is.

$$V = \int_a^b 2\pi x f(x) dx.$$

Example. Find the volume of the solid obtained by rotating about the y-axis the region bounded by $y = 2x^2 - x^3$ and $y = 0$.



Sln: Solving

$$0 = 2x^2 - x^3$$

$$0 = x^2(x - 2) \Rightarrow x = 0, \; x = 2$$

Using the cylinders. we have

$$V = \int_0^2 2\pi x f(x)\,dx = 2\pi \int_0^2 x(2x^2 - x^3)\,dx$$

$$V = 2\pi \int_0^2 (2x^3 - x^4)\,dx$$

$$V = 2\pi \left\{ \frac{x^4}{2} - \frac{x^5}{5} \Big|_0^2 \right\} = \frac{16}{5}\pi.$$

For you! try solve this problem using cross-sections perpendicular to the y-axis.